

Ria Banerjee

Full Stack Software Developer

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PROFESSIONAL SUMMARY

Backend-focused Software Developer with 3+ years of experience and double Master's degrees in CS. Skilled in Python, Go, Java, and JavaScript (React, Node), with expertise in building scalable APIs, real-time pub/sub systems (Kafka, MQTT), and AWS-based cloud architectures. Experienced in containerization, CI/CD, and integrating 20+ third-party tools to enhance system performance and reliability.

TECHNICAL SKILLS

- **Programming Languages:** Python, Go, Java, C# .Net, C++, C, Shell/Bash.
- **Web Development:** TypeScript, React, Node.js, HTML5, CSS, ElectronJS.
- **Frameworks & Tools:** Django, Flask, SpringBoot, Docker, Kubernetes, RESTful API, Kafka, MQTT, RabbitMQ, Git, Jenkins, Ansible, ServiceNow, Twilio, OAuth, ReCAPTCHA.
- **Database Technologies:** MySQL, MongoDB, PostgreSQL, SQLAlchemy, GraphQL, Redis, DynamoDB.
- **Cloud & DevOps:** GCP, AWS (EC2, EKS, ECS, Route53, S3, Lambda), Docker, Kubernetes clusters, Jenkins, CI/CD, GitOps.
- **Security and Monitoring:** Grafana, Kibana, Vault, SSL/TLS, Terraform.
- **Machine Learning:** Pandas, TensorFlow, OpenAI APIs, Data Modeling, PyTorch, Scikit-learn.

WORK EXPERIENCE

Backend Engineer, EAGL Technology Inc.

May 2024-present | Albuquerque, NM, USA

Tech Stack: Python, C/C++, Go, React, SQL, Redis, Docker, Flask, Kafka, MQTT

Project Description: Real-time gunshot detection platform deployed across high-stakes environments (schools, hospitals).

- Led the end-to-end migration of a legacy gunshot detection platform into a cloud-native microservice architecture on AWS, improving uptime to **99.99% across 100+ deployments**, including schools, public venues, city infrastructure, and law enforcement.
- **Redesigned the backend pipeline using Flask REST APIs, Celery queues, and SQLAlchemy ORM**, enabling real-time alerting and analysis with fault-tolerant failover.
- **Reduced processing latency by 70%** for MQTT & Kafka event streams by introducing Redis caching layers, async task workers, and a rate-limited Nginx reverse proxy to manage 5000+ concurrent alerts per minute.
- **Integrated an LLM-powered summarization engine** using OpenAI and LangChain to generate contextual incident summaries for admins and law enforcement in real time.
- **Implemented E911 alert workflows using Twilio**, ensuring rapid emergency dispatch during active incidents.
- Implemented E911 service using Twilio, set asynchronous message queues using Celery, implemented cache with Redis, ORM using SQLAlchemy, and managed Postgresql.
- **Automated CI/CD pipelines using Jenkins and Ansible**, reducing deployment lead time by 60% and enabling rapid iteration across environments.
- **Developed and integrated embedded communication logic** in C++ to allow sensors to transmit gunshot data via MQTT over cellular networks to cloud APIs.
- **Managed all infrastructure and backend services within AWS**, including EC2, S3, RDS (MySQL), EKS (Kubernetes), and IAM for secure role-based access.

Founding Engineer, BabelCast Inc.

Oct 2023-Jan 2024 | Chicago, IL, USA (Remote)

Tech Stack: Python, Flask, React, Node, AWS, REST, Docker, Kubernetes

Project Description: Founding engineer at a voice-based entertainment startup building real-time voice conversion during calls. Enabled users to transpose voices into 20K+ celebrity/custom models and collaborate on creations via a scalable web platform.

- Led the development of Python and Flask backend, scaling to handle **100K+ concurrent users**, deployed using Docker and Kubernetes clusters on AWS.
- Integrated **Transformer-based voice embedding** (Hugging Face), improving real-time matching accuracy by 30%.

- Incorporated over 20,000 voice models, integrated with GraphQL schema, enhancing database interactions and **reducing query response time by 50%**.
- Used WebRTC + Flask-SocketIO for real-time voice conversion pipeline; integrated S3 for blob storage and OAuth2.
- Built testing frameworks with Gatling and custom async WebSocket tests; collaborated on front-end/backend integration.

Graduate Research Assistant, UNC Charlotte

Mar 2023-Oct 2023 | Charlotte, NC, USA

Tech Stack: Python, SQL, React, TensorFlow, Machine Learning models

Project Description: Collaborated with a healthcare research group to analyze biomedical data. The goal was to develop a machine learning model capable of predicting anxiety under different conditions, with applications aimed at supporting individuals with disabilities.

- Created ML pipelines with Logistic Regression, KNN, and CNNs to achieve 80%+ accuracy in anxiety classification.
- Built a **React dashboard** with interactive visualizations for experimental researchers.
- **Experimented with LLMs for natural language tagging of symptom reports**, using OpenAI and fine-tuned BERT models to label patient feedback with 90% precision (side module).

Systems Engineer, Equifax Inc. via Tata Consultancy Services

May 2021-Aug 2022 | Remote, India

Tech Stack: Java, SpringBoot, Maven, GCP, Kubernetes, SQL, Grafana

Project Description: The project involved optimizing on-premise infrastructure and leading the cloud migration, including traffic redirection and safe decommissioning of legacy systems.

- Engineered microservices for the Equifax UK system using SpringBoot, enhancing the UI and application features, resulting in a 150% boost in user satisfaction scores.
- Part of the migration team for legacy systems to GCP migration, architecting Kubernetes clusters and configuring Grafana for monitoring, achieving 200% improvement in uptime and reliability.
- Implemented CI/CD practices using Jenkins and ServiceNow for microservice deployments, achieving a 95% release success.
- Conducted code quality checks and vulnerability management using Nexus Lifecycle and SonarQube.

PROJECTS

Personal Portfolio Project

Created a personal portfolio website using React+Node+Vite+Nginx (frontend) and Python+Flask (backend). Utilized Google ReCAPTCHA, integrated Rate Limiting, SSL certs, and reverse proxy. Dockerized and deployed to AWS (EC2+ECS).

Large Language Model (LLM) based Chatbot in Python

Created a Chatbot trained on personal resume using **Retrieval-Augmented Generation (RAG)** integrated with Django-based web-app, creating REST endpoints, with React, hosted on DigitalOcean.

PUBLICATION

Recommendation Systems based on Collaborative Filtering using Autoencoders: Issues and Opportunities, published in Recent Innovations in Computing, Springer. DOI: https://doi.org/10.1007/978-981-15-8297-4_32.

EDUCATION

University of North Carolina at Charlotte, USA | *Master's in Computer Science*

Aug 2022- May 2024

Nirma University, India | *Master's of Computer Application*

Aug 2019- May 2021

Ahmedabad University, India | *Bachelor of Computer Application*

Aug 2016- May 2019